31.4: Fungi have radiated into a diverse set of lineages

In the past decade, molecular analyses have reshaped our understanding of the evolutionary relationships between fungal groups. In addition, metagenomic studies have led to the discovery of entirely new groups of fungi. As a result, the phylogeny of fungi is undergoing dramatic change. For example, one traditional group, the Zygomycota, has been abandoned because it was paraphyletic, and its members have been reassigned to other groups. Recent studies also indicate that the *microsporidians*, an enigmatic group of unicellular parasites, should be classified as fungi and may belong to a basal fungal lineage (one that diverged from other fungi early in the history of the group).

Figure 31.10 presents a current hypothesis of the relationships among fungal groups. In this section, we'll survey the groups identified in this phylogenetic tree. However, the groups shown in Figure 31.10 may represent only a small fraction of the diversity of extant fungal groups. (Extant lineages are those that have surviving members.) While there are about 145,000 known species of fungi, in recent years more than 2,000 new species have been discovered annually. By some estimates the actual number of fungal species lies between 2.2 and 3.8 million—more than all of the 1.9 million species of organisms (of every type) that biologists have currently identified and named.

Figure 31.10

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Phylogeny of fungi.
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An evolutionary tree.

This phylogenetic hypothesis shows major clades of extant fungi; dotted lines indicate evolutionary relationships that are uncertain.

The tree begins with an ancestral protist. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids, while branch 5 splits into branches 6 (Zoopagomycetes) and 7. Branch 7 splits into branches 8 (Mucoromycetes) and branch 9. Branch 9 splits into branches 10 (Ascomycetes) and 11 (Basidiomycetes).

Cryptomycetes and Microsporidians

An evolutionary tree highlights cryptomycetes and microsporidians.

The tree begins with a node splitting into various branches. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. These two branches are highlighted. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids, while branch 5 splits into branches 6 (Zoopagomycetes) and 7. Branch 7 splits into branches 8 (Mucoromycetes) and branch 9. Branch 9 splits into branches 10 (Ascomycetes) and 11 (Basidiomycetes).

Genomic studies indicate that **cryptomycetes** (fungi in the phylum Cryptomycota) and **microsporidians** (fungi in the phylum Microsporidia) form a sister group and are a basal fungal lineage (see Figure 31.10). While most molecular comparisons support the placement of cryptomycetes and microsporidians at the base of the fungal tree, more data are needed to help resolve this phylogeny.

Cryptomycetes

Although only 30 species have been identified to date, genetic data suggest that the cryptomycetes are a large and diverse group. DNA sequences from members of this group have been found in marine and freshwater communities, as well as soils. Cryptomycetes also have been found in aerobic and anaerobic environments, and in geographical locations across the globe. Like the species shown in **Figure 31.11**, *Rozella allomycis*, many of the cryptomycetes identified to date are parasites of protists and other fungi.

Figure 31.11

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    The cryptomycete *Rozella allomycis* parasitizing another
fungus.
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Rozella allomycis is a parasitic fungi whose spores invade the hypha of the host fungus.

Cryptomycetes are unicellular and have flagellated spores. Cryptomycetes also can synthesize a chitin-rich cell wall, a key structural feature of the fungi (see Concept 31.1).

Microsporidians

The 1,300 species of microsporidians are unicellular parasites of protists and animals, including humans **(Figure 31.12)**. Infections in humans can cause reduced longevity and weight loss. The microsporidian *Nosema ceranae* is a parasite of honeybees and may contribute to *Colony Collapse Disorder*, a devastating outbreak that has led to the loss of honeybee colonies throughout the world.

Figure 31.12

A spore of the microsporidian *Fibrillanosema crangonycis*.

A spore of microsporidian, Fibrillanosema crangonycis showing an elliptical structure with smaller units along the sides and bottom edge. (scale 1 micrometer)

(TEM)

Like all fungi, microsporidians can synthesize a chitin-rich cell wall. Other aspects of their biology are unusual. For example, microsporidians have highly reduced mitochondria and small genomes, with only 2,000 genes in some species. The genome of one microsporidian, *Encephalitozoon intestinalis*, has just 2.3 Mb of DNA—the smallest genome of any eukaryote sequenced to date. Unlike other basal fungi, microsporidians lack flagellated spores; instead, they produce unique spores that infect host cells via a harpoon-like organelle.

Chytrids

An evolutionary tree highlights chytrids.

The tree begins with a node splitting into various branches. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids which is highlighted, while branch 5 splits into branches 6 (Zoopagomycetes) and 7. Branch 7 splits into branches 8 (Mucoromycetes) and branch 9. Branch 9 splits into branches 10 (Ascomycetes) and 11 (Basidiomycetes).

The fungi classified in phylum Chytridiomycota, called **chytrids**, are ubiquitous in lakes and soil; recent metagenomic studies have uncovered new clades of chytrids in hydrothermal vent and other marine communities. Some of the approximately 1,000 chytrid species are decomposers, while others are parasites of protists, other fungi, plants, or animals; as we'll see later in the chapter, two chytrid parasites have contributed to the global decline of amphibian populations. Still other chytrids are important mutualists. For example, anaerobic chytrids that live in the digestive tracts of sheep and cattle help to break down plant matter, thereby contributing significantly to the animal's growth.

Nearly all chytrids have flagellated spores, called **zoospores(Figure 31.13)**. Like other fungi, chytrids have cell walls made of chitin, and they also share certain key enzymes and metabolic pathways with other fungal groups. Some chytrids form colonies with hyphae, while others exist as single spherical cells.

Figure 31.13

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Flagellated chytrid zoospore.
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A flagellated chytrid zoospore has a flagellum, a tail-like structure. The scale is 4 micrometers.

(TEM)

Video

Phlyctochytrium Zoospore Release

image

Play

image

Zoopagomycetes

An evolutionary tree highlights zoopagomycetes.

The tree begins with a node splitting into various branches. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids, while branch 5 splits into branches 6 (Zoopagomycetes), which is highlighted and 7. Branch 7 splits into branches 8 (Mucoromycetes) and branch 9. Branch 9 splits into branches 10 (Ascomycetes) and 11 (Basidiomycetes).

Most of the 900 species of **zoopagomycetes**, fungi in the phylum Zoopagomycota, live as parasites or as commensal (neutral) symbionts of animals; some are parasites of other fungi or protists. Zoopagomycetes form filamentous hyphae and reproduce asexually by producing nonflagellated spores. Some zoopagomycetes induce insects that they parasitize to perch near the top of plants; the insects subsequently die and fungal spores are released to infect new victims **(Figure 31.14)**. Sexual reproduction, where known, involves the formation of a durable structure called a zygosporangium, which houses and protects the zygote.

Figure 31.14

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A fly covered with fungal hyphae.
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A fly hanging on the underside of a leaf, with the bottom half of its body wrapped in strips of fungal hyphae.

This fly was killed by the zoopagomycete Entomophthora muscae, also known as the "fly death fungus," for obvious reasons.

The loss of flagellated spores in the zoopagomycetes and other fungal lineages may have been associated with a transition to life on land. Basal fungal lineages had flagellated spores, enabling dispersal through water. In contrast, zoopagomycetes and all of their closest fungal relatives (the clade consisting of the mucoromycetes, ascomycetes, and basidiomycetes; see Figure 31.10) have nonflagellated spores, which are dispersed by wind in terrestrial fungi.

Mucoromycetes

An evolutionary tree highlights mucoromycetes.

The tree begins with a node splitting into various branches. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids, while branch 5 splits into branches 6 (Zoopagomycetes) and 7. Branch 7 splits into branches 8 (Mucoromycetes), which is highlighted and branch 9. Branch 9 splits into branches 10 (Ascomycetes) and 11 (Basidiomycetes).

There are approximately 750 known species of **mucoromycetes**, fungi in the phylum Mucoromycota. This phylum includes species of fast-growing molds responsible for causing foods such as bread, peaches, strawberries, and sweet potatoes to rot during storage. Although some mucoromycetes are decomposers, most are associated with plants. Many mucoromycetes live as parasites or pathogens of plants, while others live as mutualists (including some mycorrhizae).

The life cycle of *Rhizopus stolonifer* (black bread mold) is fairly typical of mucoromycete species **(Figure 31.15)**. Its hyphae spread out over the food surface, penetrate it, and absorb nutrients. The hyphae are coenocytic, with septa found only where reproductive cells are formed. In the asexual phase, bulbous black sporangia develop at the tips of upright hyphae. Within each sporangium, hundreds of genetically identical haploid spores develop and are dispersed through the air. Spores that happen to land on moist food germinate, growing into new mycelia.

Figure 31.15

The life cycle of the mucoromycete *Rhizopus stolonifer* (black bread mold).

Mycelia reproduce sexually going through plasmogamy, forming zygosporangium. Karyogamy occurs then meiosis. Zygosporangium germinates into a sporangium and disperses spores which grow into new mycelia. Mycelia can also reproduce asexually.

Sexual Reproduction. Steps are as follows.

1. Haploid mycelia have various mating types (here designated minus, with red nuclei, and +, with blue nuclei).
2. Neighboring haploid mycelia of different mating types form hyphal extensions (gametangia), each of which encloses several haploid nuclei.
3. After plasmogamy, a zygosporangium forms (heterokaryotic (n + n)), containing multiple haploid nuclei from the two parents.
4. The zygosporangium develops a rough, thick-walled coating that can resist harsh conditions for months.
5. When conditions are favorable, karyogamy occurs forming a diploid (2 n) nuclei, then meiosis.
6. The zygosporangium (haploid (n)) germinates into a sporangium on a short stalk.
7. The sporangium disperses genetically diverse haploid.
8. The spores germinate and grow into new mycelia.
9. Mycelia can also reproduce asexually by forming sporangia that produce genetically identical haploid spores.

Inset of Rhizopus growing on bread. Inset of sporangia (scale: 50 micrometers).

If environmental conditions deteriorate—for instance, if the mold consumes all its food—*Rhizopus* may reproduce sexually. The parents in a sexual union are mycelia of different mating types, which possess different chemical markers but may appear identical. Plasmogamy produces a sturdy structure called a **zygosporangium** (plural, *zygosporangia*), in which karyogamy and then meiosis occur. Note that while a zygosporangium represents the zygote (2n) stage in the life cycle, it is not a zygote in the usual sense (that is, a cell with one diploid nucleus). Rather, a zygosporangium is a multinucleate structure, first heterokaryotic with many haploid nuclei from the two parents, then with many diploid nuclei after karyogamy.

Zygosporangia are resistant to freezing and drying and are metabolically inactive. When conditions improve, the nuclei of the zygosporangium undergo meiosis, the zygosporangium germinates into a sporangium, and the sporangium releases genetically diverse haploid spores that may colonize a new substrate. Some mucoromycetes can actually “aim” and then shoot their sporangia toward bright light. **Figure 31.16** shows one example, *Pilobolus*, which decomposes animal dung. Its sporebearing hyphae bend toward light, where there are likely to be openings in the vegetation through which spores may reach fresh grass. The fungus then launches its sporangia in a jet of water that can travel up to 2.5m. Grazing animals ingest the fungi with the grass and then scatter the spores in feces, thereby enabling the next generation of fungi to grow.

Figure 31.16

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*Pilobolus* aiming its sporangia.
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Pilobolus aiming its sporangia (scale: 0.5 millimeters)

Finally, the phylum Mucoromycota also includes the **glomeromycetes**, a clade of fungi that form arbuscular mycorrhizae (see Figure 31.4b and Figure 37.14). The tips of the hyphae that push into plant root cells branch into tiny treelike arbuscules. About 85% of all plant species have mutualistic partnerships with arbuscular mycorrhizae.

Ascomycetes

An evolutionary tree highlights ascomycetes.

The tree begins with a node splitting into various branches. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids, while branch 5 splits into branches 6 (Zoopagomycetes) and 7. Branch 7 splits into branches 8 (Mucoromycetes) and branch 9. Branch 9 splits into branches 10 (Ascomycetes), which is highlighted and 11 (Basidiomycetes).

Mycologists have described 90,000 species of **ascomycetes**, fungi in the phylum Ascomycota, from a wide variety of marine, freshwater, and terrestrial habitats. The defining feature of ascomycetes is the production of spores (called ascospores) in saclike **asci** (singular, *ascus*); thus, they are commonly called *sac fungi*. During their sexual stage, most ascomycetes develop fruiting bodies, called **ascocarps**, which range in size from microscopic to macroscopic **(Figure 31.17)**. The ascocarps contain the spore-forming asci.

Figure 31.17

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Ascomycetes (sac fungi).
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Two sac fungi.

The edible ascocarp of Morchella esculenta, the tasty morel, is often found under trees in orchards. Here it grows with a white stem and pocketed cap. Tuber melanosporum is a truffle species that forms ectomycorrhizae with trees. The ascocarp grows underground and emits a strong odor. These ascocarps have been dug up and the middle one is sliced open; the middle one is lighter than the others.

*Ascomycetes vary greatly in morphology (see also*Figure 31.10*). How could you confirm that a fungus is an ascomycete?*

*For suggested answer, see*Appendix A.*

Ascomycetes vary in size and complexity from unicellular yeasts to elaborate cup fungi and morels (see Figure 31.17). They include some of the most devastating plant pathogens, which we will discuss later. However, many ascomycetes are important decomposers, particularly of plant material. More than 25% of all ascomycete species live with green algae or cyanobacteria in beneficial symbiotic associations called lichens. Some ascomycetes form mycorrhizae with plants. Many others live between mesophyll cells in leaves; some of these species release toxic compounds that help protect the plant from insects.

Although the life cycles of various ascomycete groups differ in the details of their reproductive structures and processes, we'll illustrate some common elements using the bread mold *Neurospora crassa***(Figure 31.18)**. Ascomycetes reproduce asexually by producing enormous numbers of asexual spores called **conidia** (singular, *conidium*). Unlike the asexual spores of most mucoromycetes, in most ascomycetes, conidia are not formed inside sporangia. Rather, they are produced externally at the tips of specialized hyphae called conidiophores, often in clusters or long chains, from which they may be dispersed by the wind.

Figure 31.18

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The life cycle of *Neurospora crassa*, an ascomycete.
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The life cycle of Neurospora crassa, an ascomycete.

Neurospora is a bread mold and research organism that also grows in the wild on burned vegetation.

The cycle of Neurospora crassa is as follows.

1. Ascomycete mycelia can reproduce asexually by producing pigmented
2. haploid spores (conidia). Diagram is of the branching structure of the haploid mycelium dotted with spores. The spores (drawn as small, free circles) disperse; Conidia, mating type, minus labeled. Following germination, some of the haploid spores develop into haploid hypha and then into haploid mycelia.
3. Neurospora can also reproduce sexually by producing specialized hyphae. Conidia of the opposite mating type fuse to these hyphae. Mating type, minus spore diagrammed fused to the mating type + hypha here.
4. The dikaryotic (two haploid nuclei from two different parents drawn) hyphae that result from plasmogamy produce many dikaryotic asci, two of which are shown here. Dikaryotic asci drawn as

structures on two tips of the dikaryotic hyphae.

5. Karyogamy occurs within each ascus, producing a diploid nucleus (zygote).
6. Each diploid nucleus divides by meiosis, yielding four haploid nuclei (shown within the asci).
7. Each haploid (within the asci) nucleus divides once by mitosis yielding eight nuclei. Cell walls and plasma membranes develop around the nuclei, forming haploid ascospores (LM).
8. The ascospores are discharged forcibly from the asci through an opening in the ascocarp and undergo dispersal. Germinating ascospores give rise to new mycelia.

VISUAL SKILLS *What is the ploidy of a cell in the specialized hypha shown in***STEP 2***?*

*For suggested answer, see*Appendix A*.*

Conidia may also be involved in sexual reproduction, fusing with hyphae from a mycelium of a different mating type, as occurs in *Neurospora*. Fusion of two different mating types is followed by plasmogamy, resulting in the formation of dikaryotic cells, each with two haploid nuclei representing the two parents. The cells at the tips of these dikaryotic hyphae develop into many asci. Within each ascus, karyogamy combines the two parental genomes, and then meiosis forms four genetically different nuclei. This is usually followed by a mitotic division, forming eight ascospores. The ascospores develop in and are eventually discharged from the ascocarp.

Compared to the life cycle of mucoromycetes, the extended dikaryotic stage of ascomycetes (and also basidiomycetes) provides additional opportunities for genetic recombination. In *Neurospora*, for example, many dikaryotic cells can develop into asci. The haploid nuclei in these asci fuse, and their genomes then recombine during meiosis, resulting in a multitude of genetically different offspring from one mating event (see steps 3–5 in Figure 31.18).

As described in Figure 17.2, biologists in the 1930s used *Neurospora* in research that led to the one gene–one enzyme hypothesis. Today, this ascomycete continues to serve as a model research organism. In 2003, its entire genome was published. This tiny fungus has about three-fourths as many genes as the fruit fly *Drosophila* and about half as many as a human **(Table 31.1)**. The *Neurospora* genome is relatively compact, having few of the stretches of noncoding DNA that occupy so much space in the genomes of humans and many other eukaryotes. In fact, there is evidence that *Neurospora* has a genomic defense system that prevents noncoding DNA such as transposons from accumulating.

Table 31.1

Comparison of Gene Density in *Neurospora, Drosophila*, and *Homo sapiens*

A table lists the genome size, number of genes, and gene density of Neurospora crassa, Drosophila melanogaster, and Homo sapiens.

The table is as follows:

Table 31.1 Comparison of Gene Density in Neurospora, Drosophila, and Homo sapiens
Neurospora crassa (ascomycete fungus)
Drosophila melanogaster (fruit fly)
Homo sapiens (human)

Basidiomycetes

An evolutionary tree highlights basidiomycetes.

The tree begins with a node splitting into various branches. Branch 1 splits into two branches 2 and 3, which have dotted lines. Branch 2 denotes Cryptomycetes and branch 3 denotes Microsporidians. Branch 1 splits further into branches 4 and 5. Branch 4 denotes Chytrids, while branch 5 splits into branches 6 (Zoopagomycetes) and 7. Branch 7 splits into branches 8 (Mucoromycetes) and branch 9. Branch 9 splits into branches 10 (Ascomycetes) and 11 (Basidiomycetes), which is highlighted.

About 50,000 species, including mushrooms, puffballs, and shelf fungi, are called **basidiomycetes** and are classified in the phylum Basidiomycota **(Figure 31.19)**. This phylum also includes mutualists that form mycorrhizae and two groups of destructive plant parasites: rusts and smuts. The name of the phylum derives from the **basidium** (plural, *basidia*; Latin for “little pedestals”), a cell in which karyogamy occurs, followed immediately by meiosis. The club-like shape of the basidium also gives rise to the common name *club fungus*.

Figure 31.19

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Basidiomycetes (club fungi).
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Three examples of basidiomycetes: shelf fungi, puffballs, and maiden veil fungus.

Three examples of basidiomycetes. Shelf fungi are important decomposers of wood. Puffballs emit a cloud of spores. Maiden veil fungus (Dictyphora) is a fungus with an odor like rotting meat.

Basidiomycetes are important decomposers of wood and other plant material. Of all the fungi, certain basidiomycetes are the best at decomposing the complex polymer lignin, an abundant component of wood. Many shelf fungi break down the wood of weak or damaged trees and continue to decompose the wood after the tree dies.

The life cycle of a basidiomycete usually includes a long-lived dikaryotic mycelium. As in ascomycetes, this extended dikaryotic stage provides many opportunities for genetic recombination events, in effect multiplying the result of a single mating. Periodically, in response to environmental stimuli, the mycelium reproduces sexually by producing elaborate fruiting bodies called **basidiocarps(Figure 31.20)**. The common white mushrooms in the supermarket are familiar examples of a basidiocarp.

Figure 31.20

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The life cycle of a mushroom-forming basidiomycete.
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Life cycle diagram for the basidiomycota, with haploid, dikaryotic and diploid stages, as well as the processes of sexual and asexual reproduction and plasmogamy, karyogamy and meiosis highlighted.

The cycle is as follows.

1. Two haploid mycelia of different mating types undergo plasmogamy.
2. A dikaryotic mycelium forms, growing faster than, and ultimately crowding out, the haploid parental mycelia.
3. Environmental cues such as rain or change in temperature induce the dikaryotic mycelium to form compact masses that develop into basidiocarps (mushrooms, in this case).
4. The basidiocarp gills are lined with terminal dikaryotic cells called basidia.
5. 5). Karyogamy in each basidium produces a diploid nucleus, which then undergoes meiosis.
6. Each diploid nucleus yields four haploid nuclei, each of which develops into a basidiospore (SEM).
7. When mature, the basidiospores are ejected and then dispersed by the wind.
8. In a suitable environment, the basidiospores germinate and grow into short-lived haploid mycelia.

VISUAL SKILLS *Use the diagram to determine the ploidy of a cell in the aboveground stalk of a mushroom.*

*For suggested answer, see*Appendix A.**

Animation

Life Cycle of a Mushroom

image

Play

image

By concentrating growth in the hyphae of mushrooms, a basidiomycete mycelium can erect its fruiting structures in just a few hours; a mushroom pops up as it absorbs water and as cytoplasm streams in from the dikaryotic mycelium. By this process, in some species a ring of mushrooms, popularly called a “fairy ring,” may appear literally overnight **(Figure 31.21)**. The mycelium below the fairy ring expands outward at a rate of about 30 cm per year, decomposing organic matter in the soil as it grows. Some giant fairy rings are produced by mycelia that are centuries old.

Figure 31.21

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A fairy ring.
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Fairy ring of mushrooms on a grass lawn. The diameter of the ring is roughly 20 meters.

According to legend, mushroom rings spring up where fairies have danced on a moonlit night. The text provides a biological explanation of how these rings form.

After a mushroom forms, its cap supports and protects a large surface area of dikaryotic basidia on gills. During karyogamy, the two nuclei in each basidium fuse, producing a diploid nucleus (see Figure 31.20). This nucleus then undergoes meiosis, yielding four haploid nuclei, each of which ultimately develops into a basidiospore. Large numbers of basidiospores are produced: The gills of a common white mushroom have a surface area of about 200 cm^2 and may drop a billion basidiospores, which blow away.